

MANIKANTH NAMPALLY

Boca Raton, FL

(561) 542-6494 | manikanthnampally94@gmail.com | linkedin.com/in/manikanthn
github.com/maninampally

SUMMARY

Data Engineer with 2.5 years building production-grade data infrastructure and ML-ready pipelines across AWS, Azure, and GCP, processing 10+ TB/day. Specialized in streaming architectures (Kafka, Spark Structured Streaming), lakehouse design (Delta Lake, Iceberg), data governance, and AI-powered data products. Own end-to-end data infrastructure at an AI-driven SaaS startup. M.S. Information Technology and Management, Florida Atlantic University (May 2026).

TECHNICAL SKILLS

- **Languages:** Python, PySpark, SQL, Scala, Bash, SparkSQL
- **Data Engineering:** Apache Spark, Spark Structured Streaming, Kafka, Flink, Airflow, dbt Core, Databricks, Delta Lake, Apache Iceberg, Apache Beam, Great Expectations, Medallion Architecture, Schema Registry, Event-Driven Architecture, Data Governance, Data Observability
- **AWS:** S3, Glue, EMR, Kinesis, Redshift, Athena, Lambda, SageMaker, CloudWatch, KMS, IAM, API Gateway, Terraform
- **GCP:** Pub/Sub, Dataflow, BigQuery, Cloud Storage, Cloud Monitoring, GCP Data Catalog, Terraform
- **Azure:** Data Factory, Databricks, Service Bus, Blob Storage, Azure SQL, Azure Functions
- **Databases:** Snowflake, Redshift, BigQuery, PostgreSQL, TimescaleDB, DynamoDB, MongoDB
- **ML/AI/MLOps:** MLflow, LangGraph, pgvector, XGBoost, Scikit-learn, PyTorch, TensorFlow, SageMaker Endpoints, SageMaker Feature Store, BigQuery ML, Vertex AI, FinBERT, LightGBM, Evidently AI
- **DevOps:** Docker, Kubernetes, GitHub Actions, CI/CD, Git
- **BI & Visualization:** Power BI, Tableau, Matplotlib, Seaborn, Streamlit

EXPERIENCE

Opsylux LLC

Data Engineer

Nov 2025 - Present

Boca Raton, FL (Remote)

AI-driven SaaS startup building affordable alternatives to enterprise platforms

- Architected and own end-to-end data infrastructure, building idempotent ELT pipelines in Python, Airflow, and Azure Data Factory ingesting from REST APIs, flat files, and databases into Azure SQL - reducing pipeline errors from 12% to under 2%, achieving 99%+ reliability, and cutting reporting prep time by 50% for internal teams and client-facing data products.
- Designed and implemented star schema warehouse on Azure Databricks with dbt Core transformations, GitHub Actions CI/CD, and Great Expectations data validation, reducing query load time by 30% and ensuring data quality across all downstream consumers.
- Enforced data governance standards across all pipeline stages including schema validation, data lineage tracking, and SLA monitoring - maintaining pipeline compliance and reducing data incidents for 3+ downstream analytical consumers.
- Translated business requirements into production Power BI dashboards and automated pipelines, eliminating manual reporting across internal operations and client-facing products.

LTIMindtree

Data Engineer

May 2022 - Jul 2024

Hyderabad, India

Global technology consulting firm serving Fortune 500 clients

- Built and owned Python and PySpark ETL/ELT pipelines on AWS ingesting 10+ TB/day from 20+ heterogeneous sources into Redshift, BigQuery, and Databricks, cutting report delivery from 8 hours to 3 hours across multiple client environments.
- Led end-to-end design and implementation of medallion architecture (Bronze/Silver/Gold) on Databricks Delta Lake with Great Expectations validation across 25+ datasets, reducing data defects by 50% and infrastructure costs by 40%.
- Architected and built end-to-end Spark Structured Streaming and Kafka ingestion platform layered with dbt Core transformations, cutting reporting latency from 24 hours to 9 hours and establishing streaming as the production standard across all analytics teams.
- Hardened pipeline security and observability by configuring KMS encryption, IAM controls, and CloudWatch monitoring via Terraform IaC, achieving 99% pipeline reliability and reducing manual reconciliation effort by 60%.
- Refactored 50+ SQL transformations across Redshift and BigQuery using partition pruning, CTEs, and window functions, reducing average query runtime from 4.2 minutes to 88 seconds and saving ~\$18K annually in compute costs.
- Engineered dbt Core transformation workflows across 3 analytics teams and data science leads, defining SLA requirements and accelerating KPI delivery by 50% across production datasets.

PROJECTS

Real-Time Security Analytics - Streaming Lakehouse on AWS

Dec 2025 - Feb 2026

- Architected a real-time streaming security analytics platform on AWS processing 708M+ authentication events across 11K+ users and 22K+ endpoints, designing a Kinesis ingestion layer feeding S3-backed Lambda processing with a 4-tier weighted risk scoring engine (LOW/MEDIUM/HIGH/CRITICAL) at sustained 1K–10K events/sec throughput.

- Engineered production-grade observability and reliability: sub-5-minute data freshness (P95), 99.9% availability, and 15-second end-to-end latency, provisioning all infrastructure via Terraform IaC with CloudWatch dashboards, automated alerting, and zero-downtime deployments.

Artha AI - AI-Powered Financial Intelligence Platform

Mar 2026 - Present

- Architected and own a production-grade AI financial intelligence platform orchestrating LangGraph multi-agent workflows across 10+ investor philosophy models (Buffett, Graham, Dalio), scoring 500+ equities daily via scheduled Airflow DAGs ingesting from 4 financial APIs (Polygon, SEC EDGAR, FRED, OpenInsider) with dbt Core transformations on PostgreSQL/TimescaleDB - achieving 99%+ daily data completeness and zero missed trading-day runs since launch.
- Engineered FastAPI backend with Redis caching (sub-50ms p99 latency), pgvector semantic search across SEC filings and earnings transcripts, and MLflow experiment tracking across 15+ model iterations - reducing equity research time from 35 minutes to under 10 minutes per stock.
- Designed end-to-end ML experimentation framework integrating LangGraph agent orchestration with MLflow model registry, enabling fully reproducible scoring runs, continuous agent improvement, and auditability across every investment decision.

FinSentinel - Real-Time Financial News Sentiment Platform

Apr 2026 - Present

- Architected real-time financial news sentiment pipeline on GCP ingesting 500+ daily articles via Pub/Sub and Dataflow (Apache Beam) into BigQuery, with dbt Core transformation models building ticker-level sentiment aggregations and time-series trend tables across a multi-tier data architecture.
- Fine-tuned ProsusAI/FinBERT (PyTorch) achieving 0.95 F1 on 3-class financial sentiment classification, benchmarked against TensorFlow Bidirectional LSTM (0.86 F1) and LightGBM baseline (0.74 F1) across 20+ MLflow-tracked experiments - deploying via FastAPI with Redis caching, Evidently AI drift monitoring, and automated Airflow retraining triggers on GCP.

Network Security - Phishing Detection MLOps Pipeline

Jan 2026 - Mar 2026

- Built production-grade end-to-end MLOps pipeline for network phishing detection on 11K+ records, orchestrating MongoDB ingestion, KS-test statistical drift detection, schema validation, and KNN imputation preprocessing - training and evaluating Random Forest, Gradient Boosting, AdaBoost, and Logistic Regression across 26 MLflow-tracked experiments on DagsHub, achieving 0.992 F1, 0.990 precision, and 0.994 recall.
- Deployed FastAPI inference service containerized with Docker, pushed to AWS ECR, and served on a self-hosted EC2 runner via GitHub Actions CI/CD - syncing versioned artifacts and trained models to S3 with full reproducibility via MLflow model registry across every deployment.

EDUCATION

Florida Atlantic University

Aug 2024 - May 2026

M.S., Information Technology and Management | GPA: 3.9/4.0

CERTIFICATIONS

- AWS Certified Cloud Practitioner -January 2026
- AWS Certified AI Practitioner - February 2026